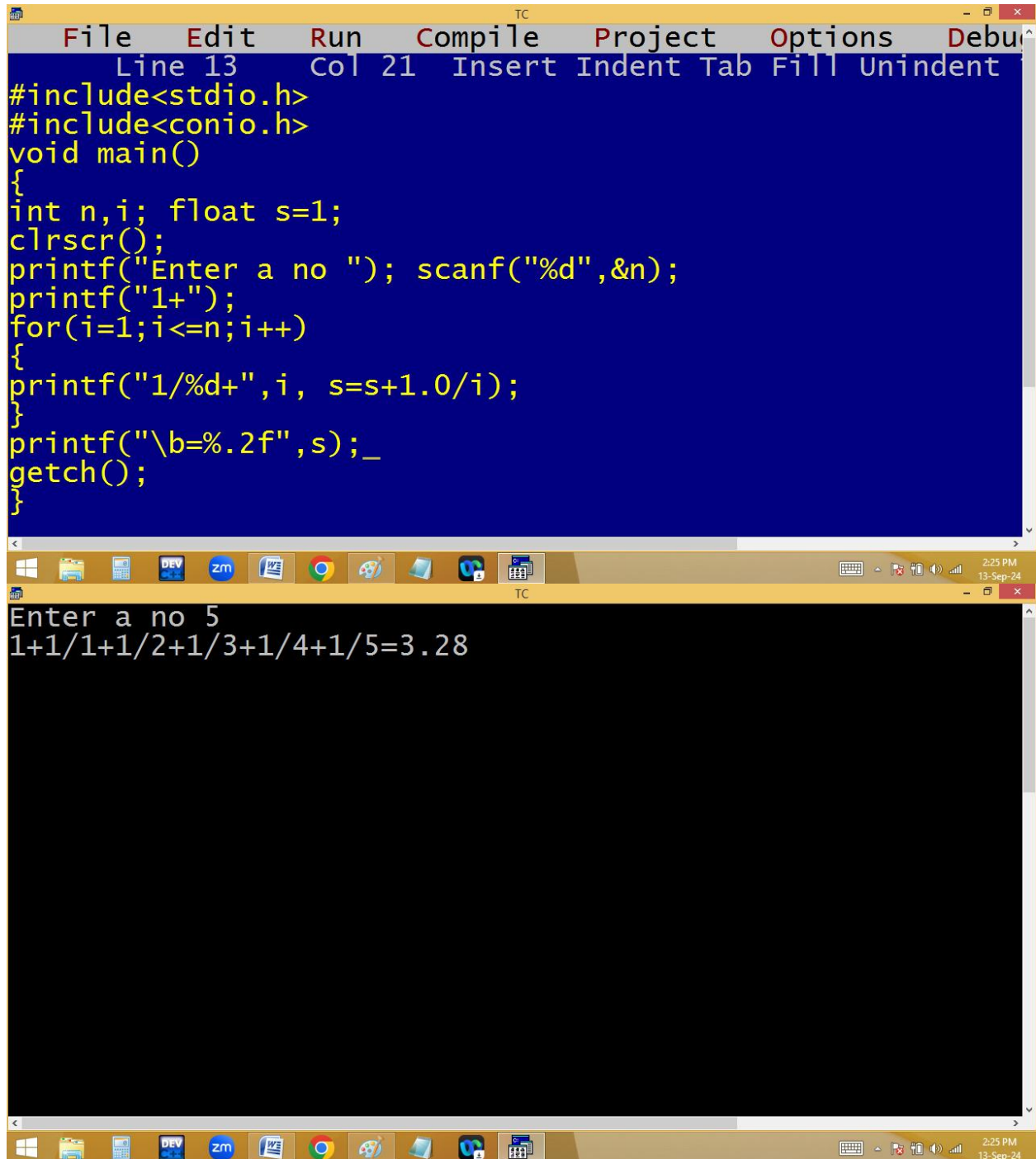


Eg: Harmonic series:

$$n=5 \rightarrow 1 + 1/1 + 1/2 + 1/3 + 1/4 + 1/5 = 3.28$$



The screenshot displays a Turbo C++ (TC) IDE window. The top menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates 'Line 13 Col 21' and provides options like Insert, Indent, Tab, Fill, and Unindent. The main editing area contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,i; float s=1;
clrscr();
printf("Enter a no "); scanf("%d",&n);
printf("1+");
for(i=1;i<=n;i++)
{
printf("1/%d+",i, s=s+1.0/i);
}
printf("\b=%.2f",s);_
getch();
}
```

Below the code editor, the program's output is shown in a separate window. It displays the prompt 'Enter a no' followed by the user input '5'. The result of the calculation is shown as '1+1/1+1/2+1/3+1/4+1/5=3.28'. The Windows taskbar at the bottom shows the time as 2:25 PM on 13-Sep-24.

$$n=5$$

$$1-2+3-4+5=3$$

```
TC
File Edit Run Compile Project Options Debug
Line 13 Col 14 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int n,i,s=0;
clrscr();
printf("Enter a no "); scanf("%d",&n);
for(i=1;i<=n;i++)
{
if(i%2==0)printf("%d+",i, s=s-i);
else printf("%d-",i,s=s+i);
}
printf("\nb=%d",s);
getch();
}

TC
Enter a no 5
1-2+3-4+5=3_
2:32 PM
13-Sep-24
```

```

for(i=1; i<=5; i++) ✓
{
    if(i%2==0) p("%d+", i, s=s-i);
    else p("%d-", i, s=s+i);
}
p("\b=%d", s);

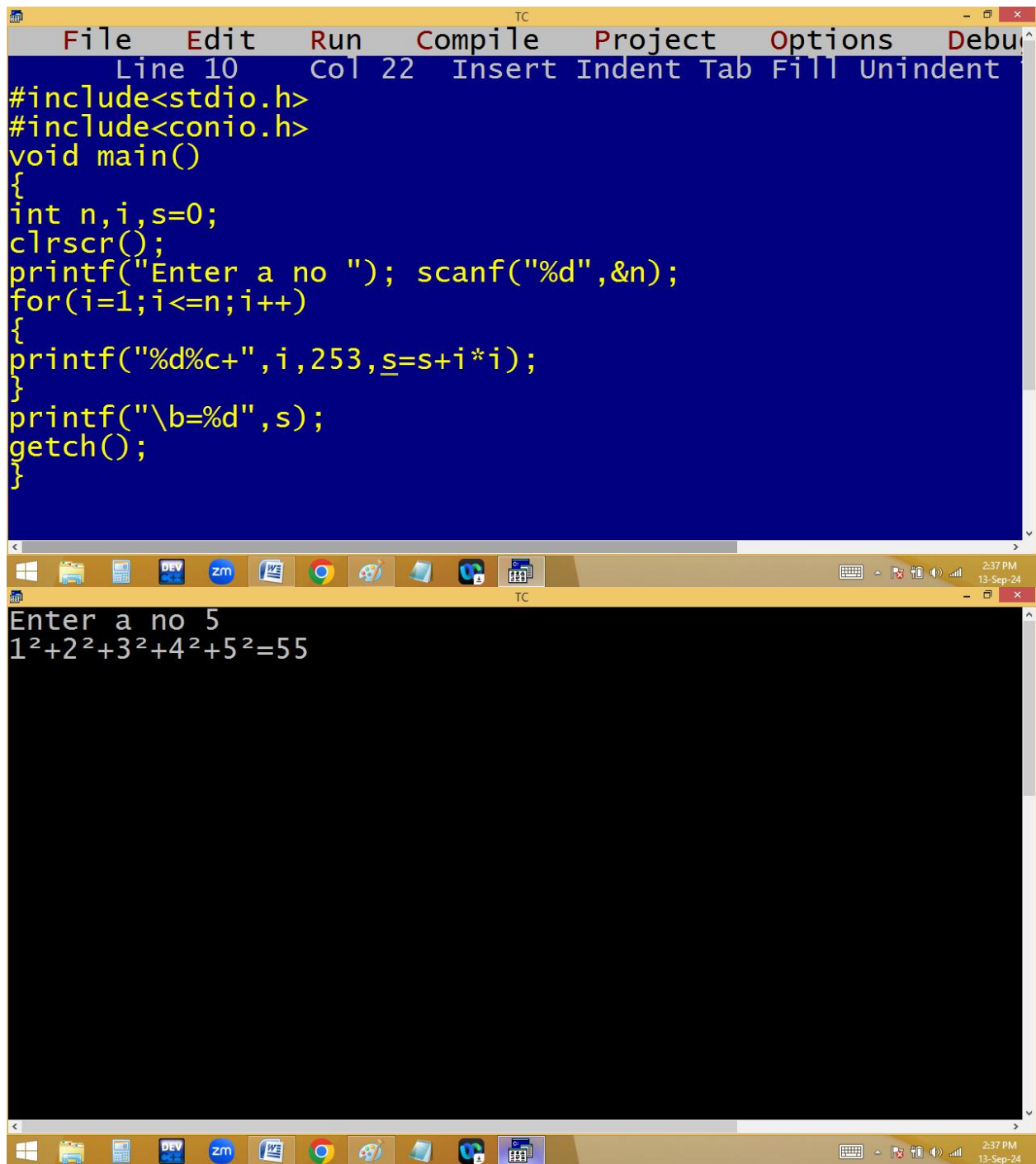
```

<u>n</u>	<u>i</u>	<u>s</u>
5	1	0+1=1
	2	1-2=-1
	3	-1+3=2
	4	2-4=-2
	5	-2+5=3
	6	

1 - 2 + 3 - 4 + 5 = 3

FINDING SQUARE SUM:

$$n=5 \rightarrow 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 1 + 4 + 9 + 16 + 25 = 55$$



```
File Edit Run Compile Project Options Debug
Line 10 Col 22 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int n,i,s=0;
clrscr();
printf("Enter a no "); scanf("%d",&n);
for(i=1;i<=n;i++)
{
printf("%d%c+",i,253,_s=s+i*i);
}
printf("\b=%d",s);
getch();
}
```

Enter a no 5
1²+2²+3²+4²+5²=55

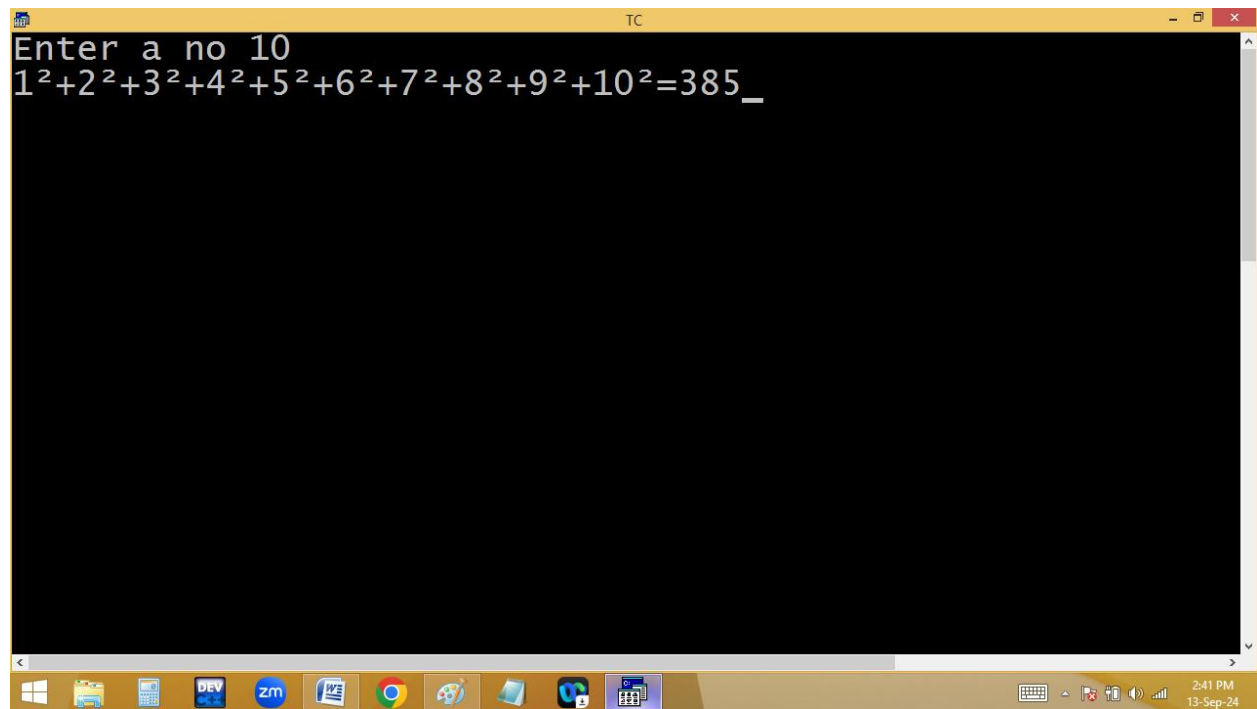
$1^2 +$
`p("%d%c+", i, 253, s=s+i*i);`



Without using \b:

```
TC
File Edit Run Compile Project Options Debug
Line 12 Col 30 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int n,i,s=0;
clrscr();
printf("Enter a no "); scanf("%d",&n);
for(i=1;i<=n;i++)
{
s=s+i*i;
if(i<n)printf("%d%c+",i,253);
else printf("%d%c=%d",i,253,s);
}
getch();
}
```

Enter a no 5
1²+2²+3²+4²+5²=55_

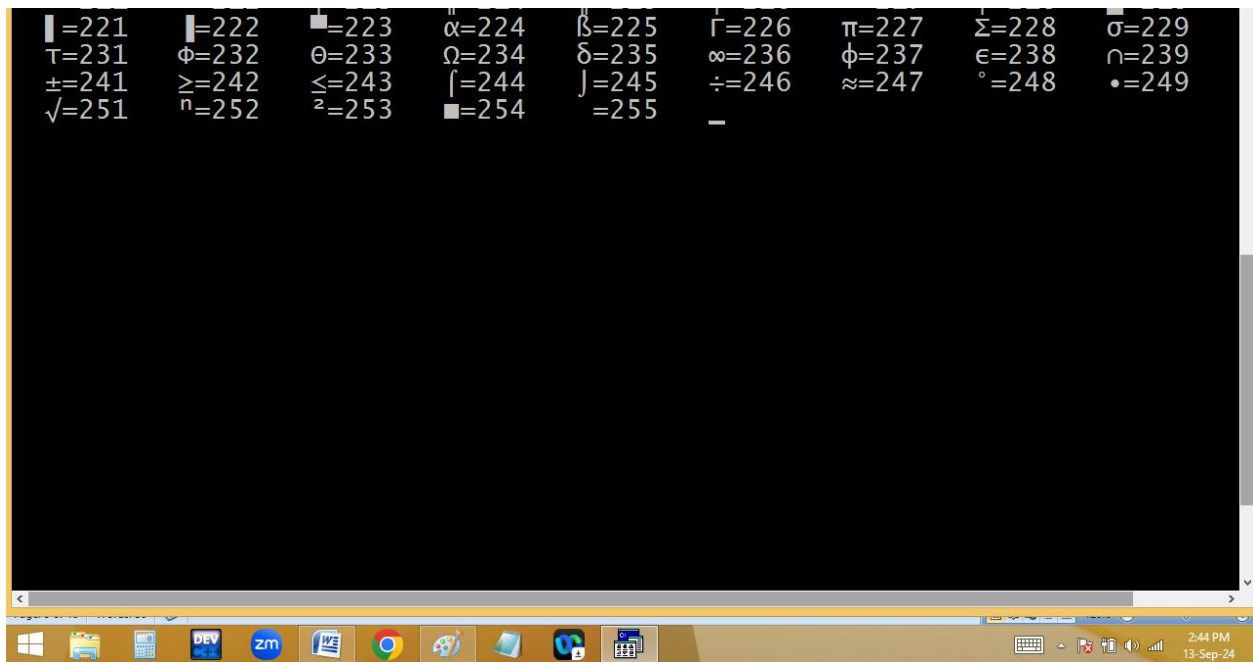


The image shows a screenshot of a Turbo C++ (TC) window. The window has a yellow title bar with the text "TC" in the center. The main area of the window is black and contains the following text in white: "Enter a no 10" followed by the equation $1^2+2^2+3^2+4^2+5^2+6^2+7^2+8^2+9^2+10^2=385_$. The window is positioned on a Windows desktop with a taskbar at the bottom. The taskbar contains several icons, including the Windows logo, a folder, a calculator, a terminal, a ZOOM icon, a Word document, a Chrome browser, a paint application, a file explorer, and a calendar. The system tray on the right side of the taskbar shows the time as 2:41 PM and the date as 13-Sep-24.

ASCII table:

```
TC
File Edit Run Compile Project Options Debug
Line 8 Col 35 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int i;
clrscr();
for(i=0;i<256;i++)printf("%c=%d\t",i,i);
getch();
}
```

=0	☺=1	☹=2	♥=3	♦=4	♣=5	♠=6	=7	=8
=9								
=13	♫=14	⚙=15	▶=16	◀=17	↑=18	!!=19	¶=20	§=21
¬=22	‡=23	↑=24	↓=25	=26	↔=27	≡=28	↕=29	△=30
▼=31	=32	!=33	"=34	#=35	\$=36	%=37	&=38	'=39
)=41	*=42	+ =43	,=44	-=45	.=46	/=47	0=48	1=49
3=51	4=52	5=53	6=54	7=55	8=56	9=57	: =58	;=59
=61	>=62	?=63	@=64	A=65	B=66	C=67	D=68	E=69
G=71	H=72	I=73	J=74	K=75	L=76	M=77	N=78	O=79
Q=81	R=82	S=83	T=84	U=85	V=86	W=87	X=88	Y=89
[=91	\=92	=93	^=94	_ =95	`=96	a=97	b=98	c=99
e=101	f=102	g=103	h=104	i=105	j=106	k=107	l=108	m=109
o=111	p=112	q=113	r=114	s=115	t=116	u=117	v=118	w=119
y=121	z=122	{=123	=124	}=125	~=126	△=127	Ç=128	ü=129
â=131	ä=132	à=133	â=134	ç=135	ê=136	ë=137	è=138	ì=139
ï=141	Ä=142	Å=143	É=144	æ=145	Æ=146	ô=147	ö=148	ò=149
ù=151	ÿ=152	Ö=153	Ü=154	£=155	£=156	¥=157	₹=158	ƒ=159
í=161	ó=162	ú=163	ñ=164	Ñ=165	ª=166	º=167	¿=168	¬=169
½=171	¼=172	¡=173	«=174	»=175	⋮=176	⋮=177	⋮=178	↓=179
†=181	‡=182	¶=183	¶=184	¶=185	¶=186	¶=187	¶=188	¶=189
‡=191	‡=192	‡=193	‡=194	‡=195	‡=196	‡=197	‡=198	‡=199
‡=201	‡=202	‡=203	‡=204	‡=205	‡=206	‡=207	‡=208	‡=209
‡=211	‡=212	‡=213	‡=214	‡=215	‡=216	‡=217	‡=218	‡=219



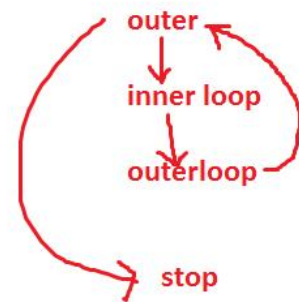
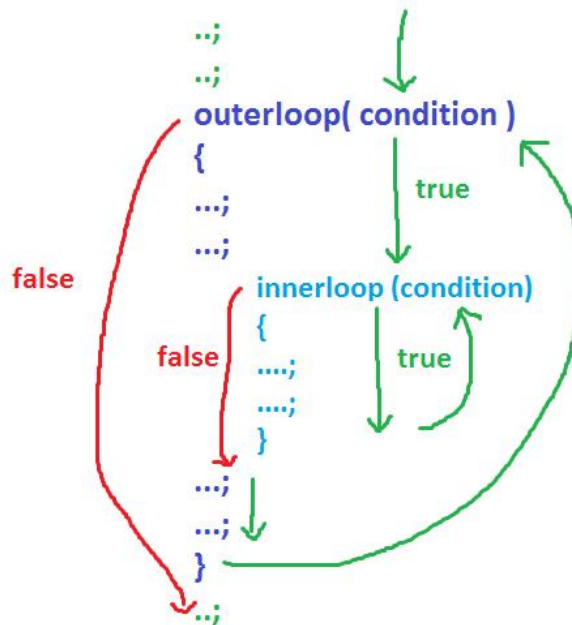
$n = 10 \rightarrow 1\ 2\ 3\ 9\ 4\ 5\ 6\ 18\ 7\ 8\ 9\ 27\ 10$

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 10 Col 46 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int i,n;
clrscr();
printf("Enter n value "); scanf("%d",&n);
for(i=1;i<=n;i++)
{
printf("%4d",i); if(i%3==0)printf("%4d",i*3);
}
getch();
}
```

Enter n value 10
1 2 3 9 4 5 6 18 7 8 9 27 10_

Nested loops:

Loop within loop is called nested loop.



Printing 1..n tables:

3 tables

1*1=1

1*10=10

2*1=2

2*10=20

3*1=3

3*10=30

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 14 Col 2 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int t,i,n;
clrscr();
printf("Enter no of tables "); scanf("%d",&n);
for(t=1;t<=n;t++)
{
for(i=1;i<=10;i++)
{
printf("%d*%d=%d\n",t,i,t*i);
}
}
getch();
}
```

Enter no of tables 2

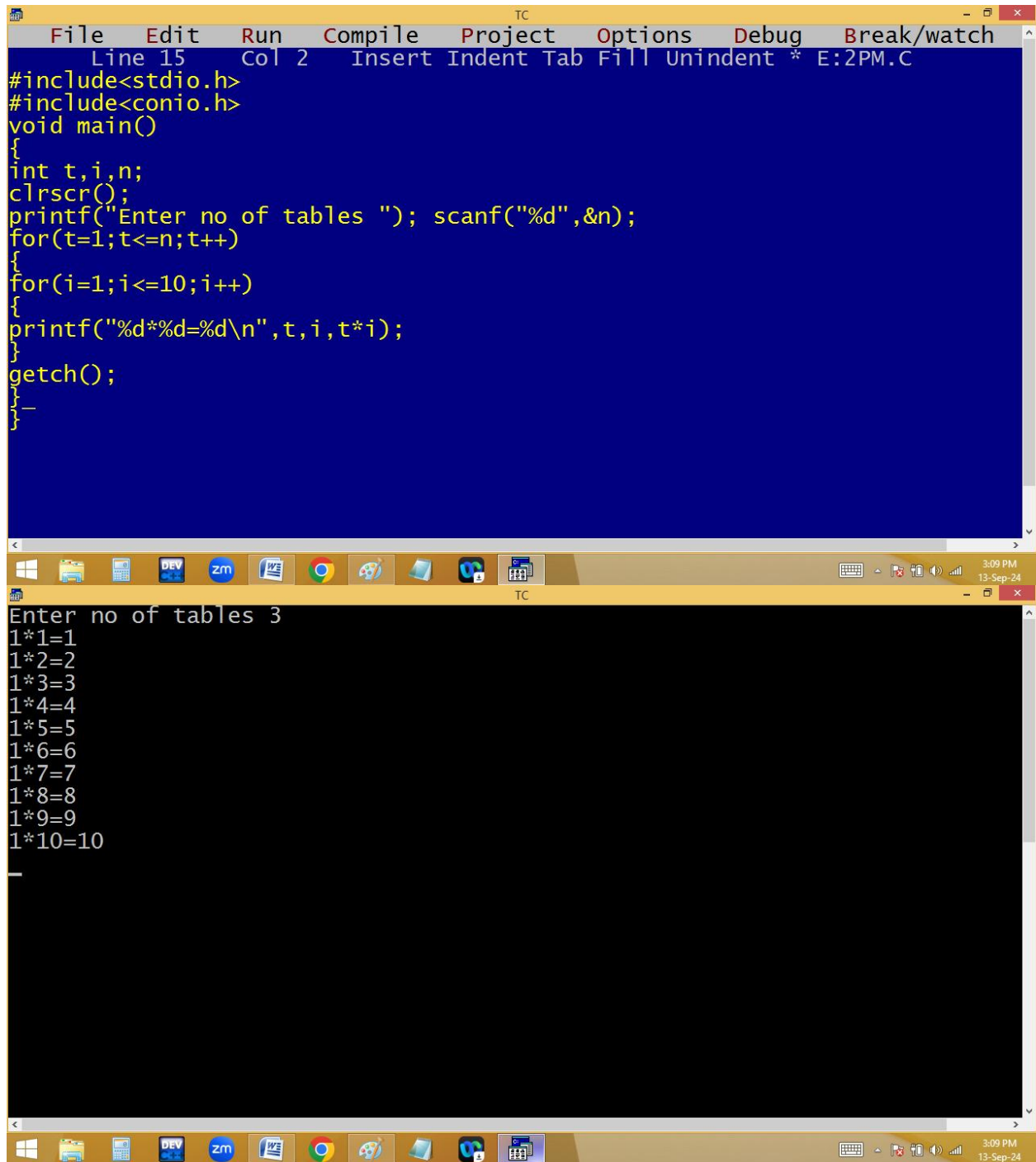
```
1*1=1
1*2=2
1*3=3
1*4=4
1*5=5
1*6=6
1*7=7
1*8=8
1*9=9
1*10=10
2*1=2
2*2=4
2*3=6
2*4=8
2*5=10
2*6=12
2*7=14
2*8=16
2*9=18
2*10=20
```

```

for( t=1; t<=3 ; t++ )
{
  for( i=1; i<=10; i++ )
  {
    p( t*i);
  }
}

```

$\frac{t}{1}$	$\times \frac{i}{1}$
	$2=2$
	10
	11
2	$\times 1=2$
	$10=20$
	11
3	$\times 1=3$
	$10=30$



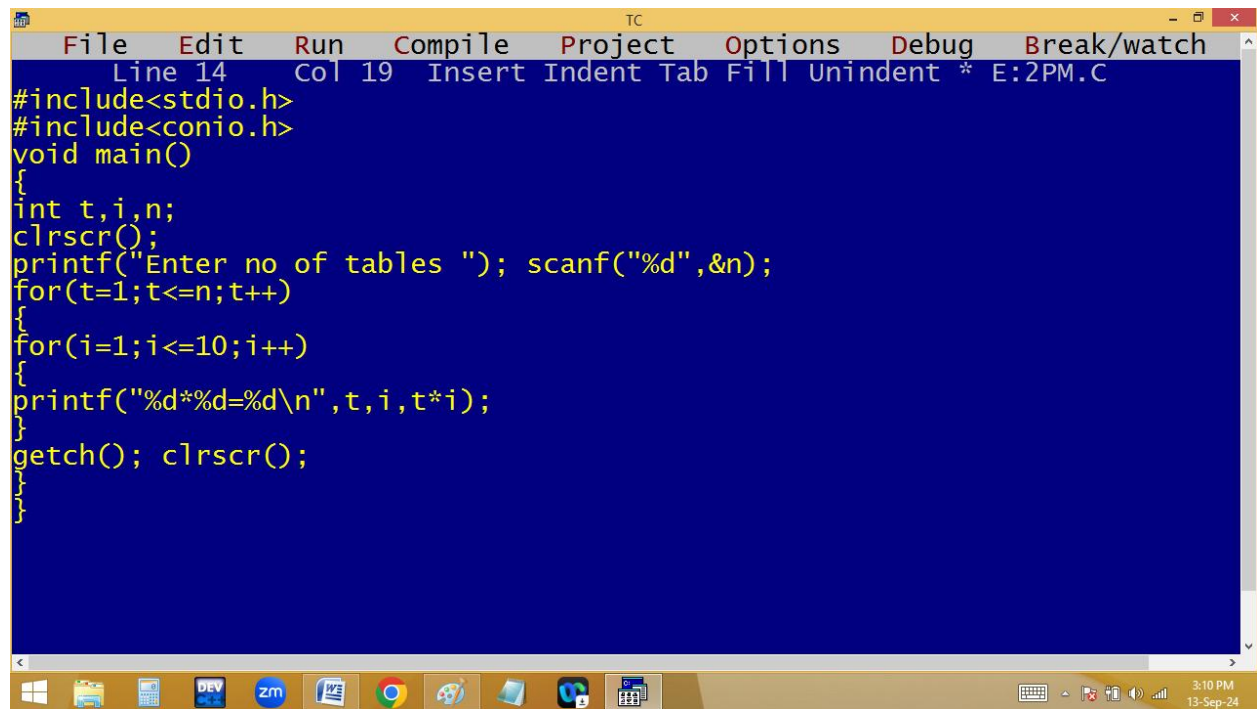
```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 15 Col 2 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int t,i,n;
clrscr();
printf("Enter no of tables "); scanf("%d",&n);
for(t=1;t<=n;t++)
{
for(i=1;i<=10;i++)
{
printf("%d*%d=%d\n",t,i,t*i);
}
}
getch();
}
```

Enter no of tables 3

```
1*1=1
1*2=2
1*3=3
1*4=4
1*5=5
1*6=6
1*7=7
1*8=8
1*9=9
1*10=10
```

```
TC
Enter no of tables 3
1*1=1
1*2=2
1*3=3
1*4=4
1*5=5
1*6=6
1*7=7
1*8=8
1*9=9
1*10=10
2*1=2
2*2=4
2*3=6
2*4=8
2*5=10
2*6=12
2*7=14
2*8=16
2*9=18
2*10=20

TC
1*9=9
1*10=10
2*1=2
2*2=4
2*3=6
2*4=8
2*5=10
2*6=12
2*7=14
2*8=16
2*9=18
2*10=20
3*1=3
3*2=6
3*3=9
3*4=12
3*5=15
3*6=18
3*7=21
3*8=24
3*9=27
3*10=30
```



The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/watch". The status bar at the top indicates "Line 14", "Col 19", and "Insert Indent Tab Fill Unindent * E:2PM.C". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int t,i,n;
    clrscr();
    printf("Enter no of tables "); scanf("%d",&n);
    for(t=1;t<=n;t++)
    {
        for(i=1;i<=10;i++)
        {
            printf("%d*%d=%d\n",t,i,t*i);
        }
        getch(); clrscr();
    }
}
```

The Windows taskbar is visible at the bottom, showing icons for various applications including a file explorer, a terminal, and a web browser. The system clock in the bottom right corner shows "3:10 PM" and "13-Sep-24".

```
Enter no of tables 2
1*1=1
1*2=2
1*3=3
1*4=4
1*5=5
1*6=6
1*7=7
1*8=8
1*9=9
1*10=10

2*1=2
2*2=4
2*3=6
2*4=8
2*5=10
2*6=12
2*7=14
2*8=16
2*9=18
2*10=20
```

Tables side by side:

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 17 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int t,i,n;
clrscr();
printf("Enter no of tables "); scanf("%d",&n);
for(i=1;i<=10;i++)
{
for(t=1;t<=n;t++)
{
printf("%d*%d=%d\t",t,i,t*i);
}
printf("\n");
}
getch();
}
```

Enter no of tables 9

1*1=1	2*1=2	3*1=3	4*1=4	5*1=5	6*1=6	7*1=7	8*1=8	9*1=9
1*2=2	2*2=4	3*2=6	4*2=8	5*2=10	6*2=12	7*2=14	8*2=16	9*2=18
1*3=3	2*3=6	3*3=9	4*3=12	5*3=15	6*3=18	7*3=21	8*3=24	9*3=27
1*4=4	2*4=8	3*4=12	4*4=16	5*4=20	6*4=24	7*4=28	8*4=32	9*4=36
1*5=5	2*5=10	3*5=15	4*5=20	5*5=25	6*5=30	7*5=35	8*5=40	9*5=45
1*6=6	2*6=12	3*6=18	4*6=24	5*6=30	6*6=36	7*6=42	8*6=48	9*6=54
1*7=7	2*7=14	3*7=21	4*7=28	5*7=35	6*7=42	7*7=49	8*7=56	9*7=63
1*8=8	2*8=16	3*8=24	4*8=32	5*8=40	6*8=48	7*8=56	8*8=64	9*8=72
1*9=9	2*9=18	3*9=27	4*9=36	5*9=45	6*9=54	7*9=63	8*9=72	9*9=81
1*10=10	2*10=20	3*10=30	4*10=40	5*10=50	6*10=60	7*10=70	8*10=80	9*10=90

```

for( i=1; i<=10; i++ )
{
  for( t=1; t<=3; t++ )
  {
    p(t*i\t);
  }
  p("\n"); ✓
}

```

<u>t</u>				<u>i</u>		
1	2	3 ✗	×	1		
1	2	3 ✗	×	2		
				3		
				⋮		
				10		
1 × 1 = 1			—	2 × 1 = 2		
1 × 2 = 2				3 × 1 = 3		
	2 × 2 = 4				3 × 2 = 6	

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 14 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int t,i,n;
clrscr();
printf("Enter no of tables "); scanf("%d",&n);
for(t=1;t<=n;t++)
{
for(i=1;i<=10;i++)
{
printf("%4d",t*i);
}
printf("\n");
}
getch();
}
```

Enter no of tables 12

1	2	3	4	5	6	7	8	9	10
2	4	6	8	10	12	14	16	18	20
3	6	9	12	15	18	21	24	27	30
4	8	12	16	20	24	28	32	36	40
5	10	15	20	25	30	35	40	45	50
6	12	18	24	30	36	42	48	54	60
7	14	21	28	35	42	49	56	63	70
8	16	24	32	40	48	56	64	72	80
9	18	27	36	45	54	63	72	81	90
10	20	30	40	50	60	70	80	90	100
11	22	33	44	55	66	77	88	99	110
12	24	36	48	60	72	84	96	108	120

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 5 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int t,i,n;
clrscr();
printf("Enter no of tables "); scanf("%d",&n);
t=1;
while(t<=n)
{
for(i=1;i<=10;i++)
{
printf("%4d",t*i);
}
printf("\n");
t++;
}
getch();
}
```

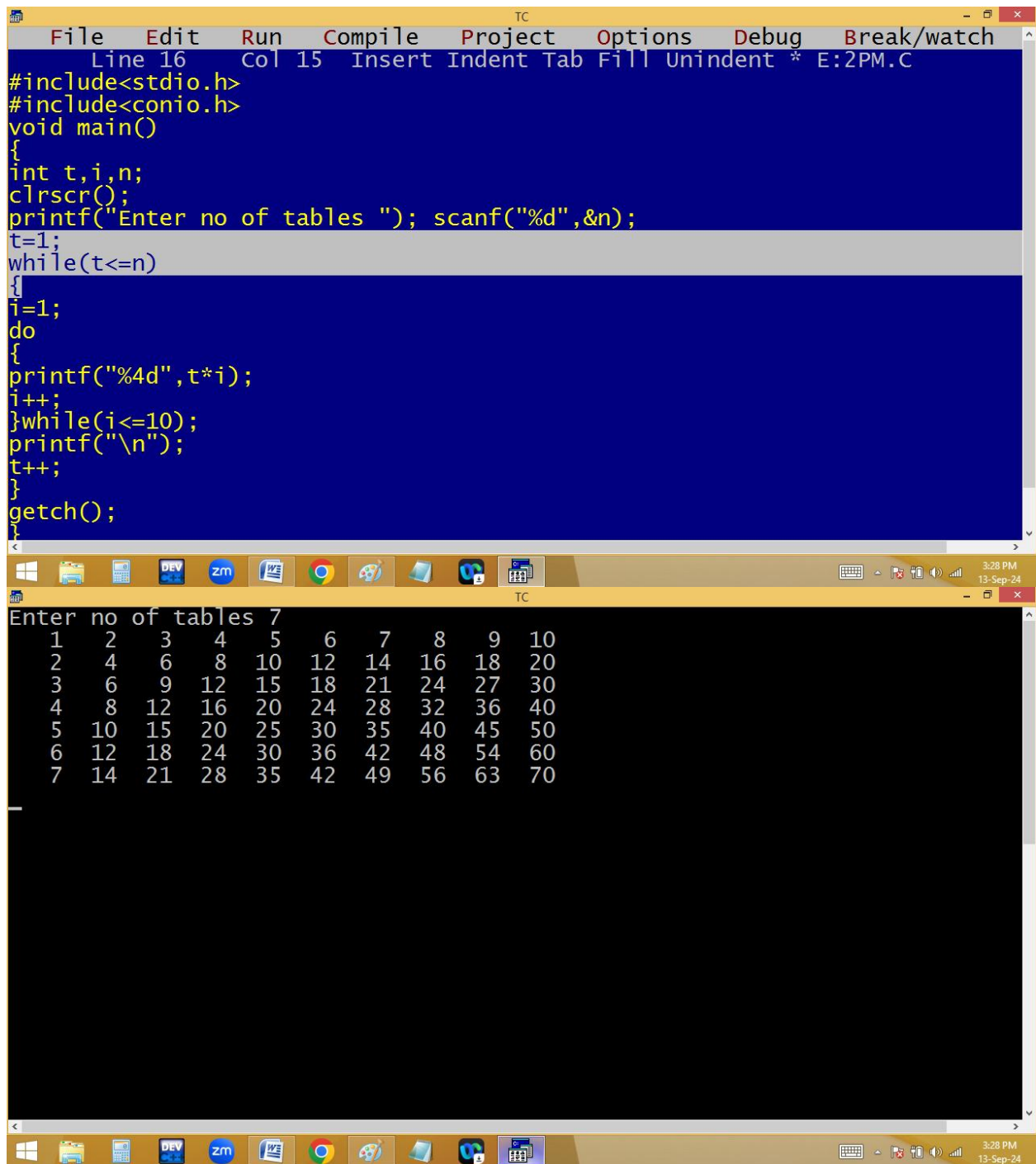
Enter no of tables 5

1	2	3	4	5	6	7	8	9	10
2	4	6	8	10	12	14	16	18	20
3	6	9	12	15	18	21	24	27	30
4	8	12	16	20	24	28	32	36	40
5	10	15	20	25	30	35	40	45	50

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int t,i,n;
clrscr();
printf("Enter no of tables "); scanf("%d",&n);
t=1;
while(t<=n)
{
i=1;
while(i<=10)
{
printf("%4d",t*i);
i++;
}
printf("\n");
t++;
}
getch();
}
```

Enter no of tables 2

1	2	3	4	5	6	7	8	9	10
2	4	6	8	10	12	14	16	18	20



```
File Edit Run Compile Project Options Debug Break/watch
Line 16 Col 15 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int t,i,n;
clrscr();
printf("Enter no of tables "); scanf("%d",&n);
t=1;
while(t<=n)
{
i=1;
do
{
printf("%4d",t*i);
i++;
}while(i<=10);
printf("\n");
t++;
}
getch();
}
```

Enter no of tables 7

1	2	3	4	5	6	7	8	9	10
2	4	6	8	10	12	14	16	18	20
3	6	9	12	15	18	21	24	27	30
4	8	12	16	20	24	28	32	36	40
5	10	15	20	25	30	35	40	45	50
6	12	18	24	30	36	42	48	54	60
7	14	21	28	35	42	49	56	63	70

Finding generic root of given no

5677 = 5+6+7+7=25 → 2+5=7

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 16 Col 30 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
long n, s=0;
clrscr();
printf("Enter the no "); scanf("%ld",&n);
while(n>9)
{
for(s=0; n!=0; n=n/10 )
{
s=s+n%10;
}
n=s;
}
printf("Generic root=%ld",n);_
getch();
}
```

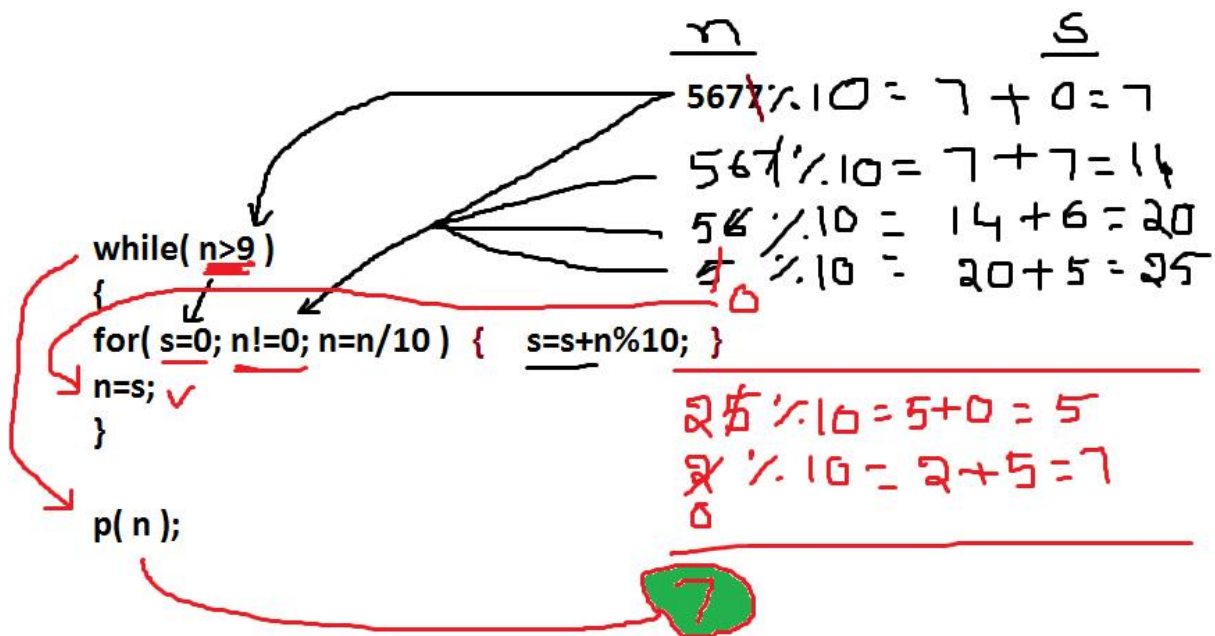
Enter the no 5677
Generic root=7

TC

```
TC
Enter the no 1689
Generic root=6
```

```
TC
Enter the no 123456789
Generic root=9
```

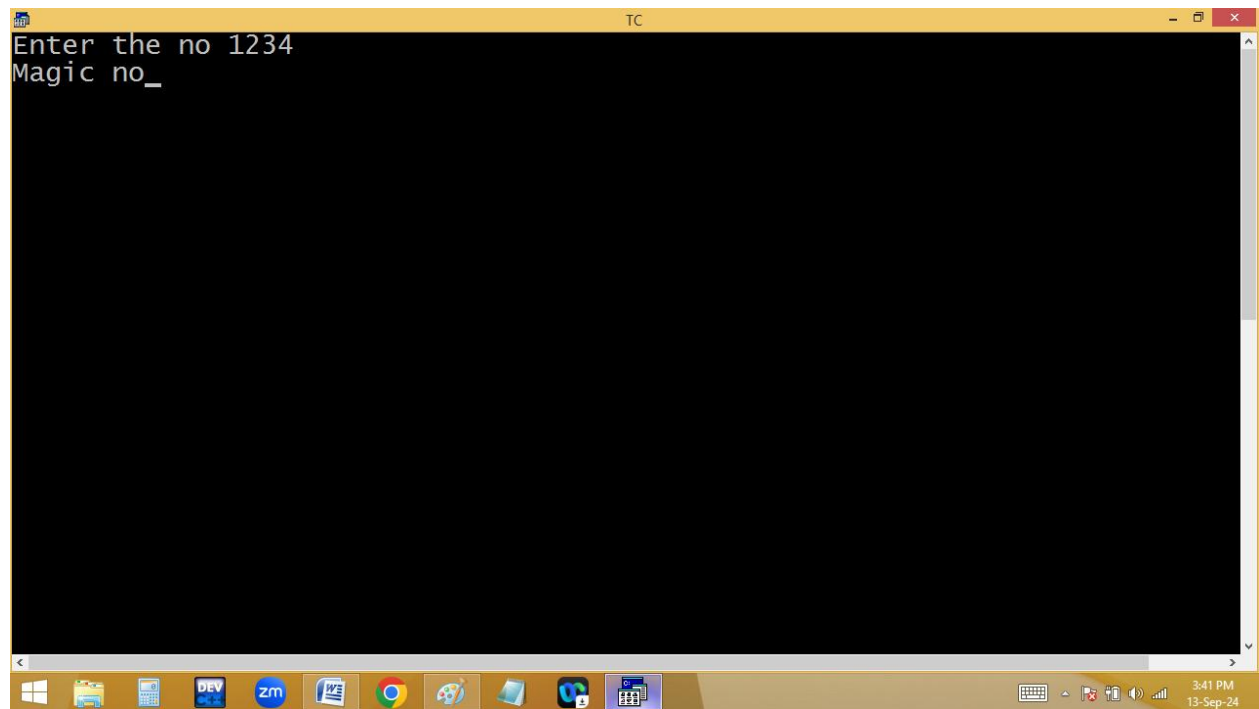
```
Enter the no 1234
Generic root=1
```



Finding magic no? when generic root is 1 it is as magic no

```
File Edit Run Compile Project Options Debug Break/watch
Line 16 Col 57 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
long n, s=0;
clrscr();
printf("Enter the no "); scanf("%ld",&n);
while(n>9)
{
for(s=0; n!=0; n=n/10 )
{
s=s+n%10;
}
n=s;
}
if(n==1) printf("Magic no");else puts("Not a Magic no");
getch();
}
```

Enter the no 2345
Not a Magic no

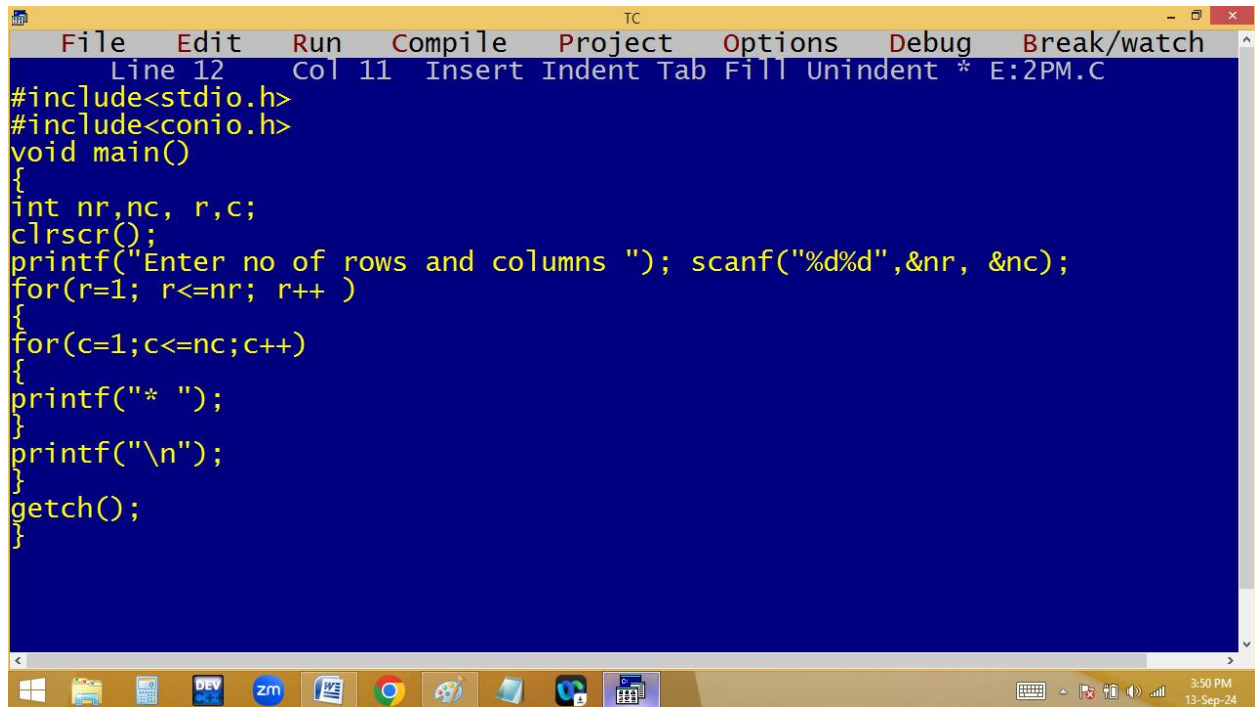


The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot displays the source code for a C program that prints a star pattern. The code is as follows:

```
File Edit Run Compile Project Options Debug Break/watch
Line 16 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc, r,c;
clrscr();
printf("Enter no of rows and columns "); scanf("%d%d",&nr, &nc);
for(r=1; r<=nr; r++ )
{
for(c=1;c<=nc;c++)
{
printf("*");
}
printf("\n");
}
getch();
}
```

The bottom screenshot shows the execution of the program. The user has entered '3' for rows and '4' for columns. The output is a 3x4 grid of stars:

```
Enter no of rows and columns 3 4
****
****
****
```



The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/watch". The status bar at the top indicates "Line 12", "Col 11", and "Insert Indent Tab Fill Unindent * E:2PM.C". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int nr,nc, r,c;
    clrscr();
    printf("Enter no of rows and columns "); scanf("%d%d",&nr, &nc);
    for(r=1; r<=nr; r++ )
    {
        for(c=1;c<=nc;c++)
        {
            printf("* ");
        }
        printf("\n");
    }
    getch();
}
```

The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, Calculator, DEV C++, Zm, Word, Chrome, Paint, and other applications. The system clock in the bottom right corner shows "3:50 PM" and "13-Sep-24".

```

Enter no of rows and columns 10 10
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *

```

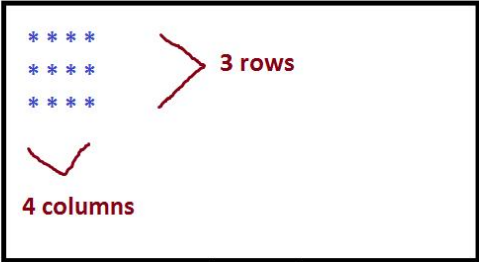
```

for( r=1; r<=3; r++ )
{
for( c=1; c<=4; c++ )
{
printf("*");
}
p("\n");
}

```

$r \leq 3$
1
2
3
~~4~~

$c = 1 \leq 4$
1 2 3 4
* * * *
1 2 3 4
* * * *
1 2 3 4
* * * *



[illegible]

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 15 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc, r,c;
clrscr();
printf("Enter no of rows and columns "); scanf("%d%d",&nr, &nc);
for(r=1; r<=nr; r++)
{
for(c=1;c<=nc;c++)
{
printf("%3d",r);
}
printf("\n");
}
getch();
}
```

Enter no of rows and columns 10 10

```
1 1 1 1 1 1 1 1 1 1
2 2 2 2 2 2 2 2 2 2
3 3 3 3 3 3 3 3 3 3
4 4 4 4 4 4 4 4 4 4
5 5 5 5 5 5 5 5 5 5
6 6 6 6 6 6 6 6 6 6
7 7 7 7 7 7 7 7 7 7
8 8 8 8 8 8 8 8 8 8
9 9 9 9 9 9 9 9 9 9
10 10 10 10 10 10 10 10 10 10
```

1 1 1 1
1 1 2 2
1 3 3 3
4 4 4 4